

TEKAMAH, NE, USA

WMO: 725527

Lat: 41.764N

Lon: 96.178W

Elev: 313

StdP: 97.62

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 94978

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.1	-18.0	-25.0	0.4	-20.4	-22.5	0.5	-18.0	15.5	-5.9	14.1	-5.1	3.3	320	0.592

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	34.5	24.1	32.9	23.7	31.5	23.1	26.5	31.6	25.5	30.8	24.5	29.6	6.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.1	21.0	29.8	23.9	19.5	28.8	22.8	18.2	27.7	84.8	31.5	80.2	30.9	75.9	29.5	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.8	11.4	10.2	DB	-26.1	37.4	2.2	1.7	-27.6	38.6	-28.9	39.6	-30.1	40.6	-31.7	41.8	(4)
(5)			WB	-26.3	28.4	2.2	0.8	-27.9	29.0	-29.1	29.4	-30.3	29.9	-31.9	30.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	-5.5	-3.3	3.7	10.6	16.9	22.4	24.5	22.6	18.7	11.2	3.6	-3.4	(6)	
	DBStd	11.79	6.78	6.95	6.61	5.53	4.90	3.69	3.19	3.00	4.48	5.36	5.68	6.37	(7)	
	HDD10.0	1797	479	374	214	58	4	0	0	0	1	49	202	416	(8)	
	HDD18.3	3529	738	607	453	239	87	8	1	3	51	228	441	674	(9)	
	CDD10.0	1885	0	1	20	74	219	373	448	391	261	87	11	0	(10)	
	CDD18.3	577	0	0	1	6	43	131	192	136	60	8	0	0	(11)	
	CDH23.3	6096	0	0	19	119	502	1384	2034	1249	669	114	4	0	(12)	
	CDH26.7	2424	0	0	4	38	187	567	896	459	241	32	1	0	(13)	
(14) Wind	WSAvg	4.5	4.8	4.9	5.3	5.7	5.3	4.5	3.3	3.0	3.5	4.3	4.7	4.6	(14)	
(15) Precipitation	PrecAvg	721	17	19	51	75	104	103	80	85	80	52	34	19	(15)	
	PrecMax	1064	58	70	172	213	286	277	215	233	196	178	112	68	(16)	
	PrecMin	379	0	0	0	10	33	19	20	9	4	0	0	1	(17)	
	PrecStd	164	12	15	37	49	49	53	40	54	47	40	29	13	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.7	17.4	26.0	31.0	34.1	36.3	37.1	35.4	33.9	30.4	22.9	14.1	(19)	
		MCWB	7.0	10.8	14.7	17.5	20.7	23.7	25.4	24.7	22.2	19.9	14.1	8.9	(20)	
	2%	DB	9.0	12.3	21.8	27.0	31.2	33.7	34.8	32.7	32.0	26.5	19.0	10.6	(21)	
		MCWB	4.7	8.0	13.8	16.2	20.0	23.1	25.2	25.0	22.0	17.4	12.1	6.7	(22)	
	5%	DB	5.9	8.7	17.6	23.4	28.6	32.1	32.9	31.1	29.4	23.6	15.9	7.6	(23)	
		MCWB	2.8	5.6	12.0	14.2	18.5	22.2	25.1	24.3	21.2	15.9	10.2	4.5	(24)	
	10%	DB	3.1	5.8	14.0	20.7	26.1	30.1	31.4	29.1	27.4	21.1	12.7	4.9	(25)	
		MCWB	1.2	3.2	9.1	13.0	17.4	21.6	24.3	23.0	19.8	14.3	8.2	2.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.6	12.1	17.4	18.9	23.5	26.4	28.5	27.3	25.0	21.5	15.7	10.9	(27)	
		MCDB	13.3	15.3	23.0	27.2	30.2	32.4	33.1	31.7	30.8	27.8	20.6	12.7	(28)	
	2%	WB	5.1	7.9	15.1	17.2	21.6	25.0	27.1	25.9	23.6	19.2	13.2	6.8	(29)	
		MCDB	8.9	11.4	19.2	24.4	28.2	30.9	31.9	30.9	29.0	23.9	18.1	9.7	(30)	
	5%	WB	3.0	5.5	12.3	15.4	20.1	24.0	25.9	25.0	22.5	17.2	10.8	4.7	(31)	
		MCDB	5.3	8.7	17.6	22.2	26.2	29.7	31.1	29.8	27.5	21.7	14.9	7.3	(32)	
	10%	WB	1.5	3.4	9.5	13.7	18.6	22.9	25.0	23.9	21.3	15.1	8.4	2.8	(33)	
		MCDB	3.1	5.4	14.2	19.5	24.2	28.6	30.3	28.1	25.9	20.1	12.6	4.9	(34)	
(35) Mean Daily Temperature Range	MDBR		10.5	10.5	12.0	13.3	12.5	12.1	11.9	11.9	14.0	13.7	12.2	10.0	(35)	
	5% DB	MCDBR	14.7	14.5	18.3	18.7	16.3	14.2	13.3	13.8	16.1	17.7	17.6	14.1	(36)	
		MCWBR	10.9	10.3	11.0	9.9	7.2	6.2	6.3	7.0	7.5	9.3	11.1	10.5	(37)	
	5% WB	MCDBR	13.4	13.7	16.4	15.6	13.5	11.9	11.8	12.0	13.2	14.3	16.0	13.0	(38)	
		MCWBR	10.2	10.0	10.2	9.1	7.0	6.4	6.4	6.8	6.9	8.6	10.7	10.0	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.288	0.314	0.335	0.395	0.419	0.422	0.432	0.422	0.380	0.334	0.299	0.288	(40)	
	taud		2.417	2.366	2.378	2.227	2.229	2.284	2.253	2.281	2.356	2.472	2.494	2.438	(41)	
	Ebn at Noon		865	892	912	874	859	855	841	838	852	859	849	833	(42)	
	Edh at Noon		78	96	108	135	138	131	134	127	109	86	72	71	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.97	2.95	3.80	4.88	5.76	6.42	6.53	5.57	4.57	3.18	2.21	1.64	(44)	
	RadStd		0.18	0.19	0.33	0.36	0.43	0.49	0.40	0.39	0.29	0.38	0.14	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page